Prob.6 (Z&T, Ch. 7, 2.4)

Use sep-n of var-s on the Laplace eq-n in $\mathbb{R}^3$ in spherical coor-s to get the separated eq-ns:

(1) $(r^2 R')' - \mu R = 0$

(2) $\Lambda_3 Y + \mu Y = 0$

where μ is a real const., & $\Lambda_3 u$ is the spherical part of the Laplace operator in $\mathbb{R}^3$:

(3) $$\Delta u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\Lambda_3 u$$

(4) $$\Lambda_3 u = \frac{1}{\sin(\varphi)}\frac{\partial}{\partial \varphi}\left[\sin(\varphi)\frac{\partial u}{\partial \varphi}\right] + \frac{1}{\sin^2(\varphi)}\cdot\frac{\partial^2 u}{\partial \theta^2}$$

Sol-n.

[Preamble].

(5) Let $u = R(r)\cdot Y(\theta,\varphi)$ for some $R(r)\neq 0$, $Y(\theta,\varphi)\neq 0$

If $R(r)=0$ or $Y(\theta,\varphi)=0 \Rightarrow u=0$, then trivially $\Delta u \equiv 0$.

$$(3)\quad \Delta u = \frac{1}{r^2}\cdot\underbrace{\left[\frac{\partial}{\partial r}\left(r^2\cdot\overbrace{\frac{\partial u}{\partial r}}^{(1)}\right)\right]}_{(2)} + \frac{1}{r^2}\cdot\underbrace{\Lambda_3 u}_{(4)} = 0.$$

$$(1)\quad \frac{\partial u}{\partial r} \overset{\text{by (5)}}{=} \frac{\partial}{\partial r}\left[R(r)\cdot Y(\theta,\varphi)\right] = R'(r)\cdot Y(\theta,\varphi)$$

$$(2)\quad \frac{\partial}{\partial r}\left[r^2\cdot R'(r)\cdot Y(\theta,\varphi)\right] = 2r\cdot R'(r)\cdot Y(\theta,\varphi) + r^2\cdot R''(r)\cdot Y(\theta,\varphi)$$

$$(3)\quad \frac{1}{r^2}\cdot(2) = \frac{1}{r^2}\cdot\left[2r\cdot R'(r)\cdot Y(\theta,\varphi) + r^2\cdot R''(r)\cdot Y(\theta,\varphi)\right] =$$

$$= \frac{2}{r}\cdot R'(r)\cdot Y(\theta,\varphi) + R''(r)\cdot Y(\theta,\varphi).$$

$$(4)\quad \Lambda_3 u = \frac{1}{\sin\varphi}\cdot\frac{\partial}{\partial\varphi}\left[\sin(\varphi)\cdot\frac{\partial}{\partial\varphi}\overset{(5)}{[u]}\right] + \frac{1}{\sin^2(\varphi)}\cdot\frac{\partial^2 u}{\partial\theta^2} =$$

$$= \frac{1}{\sin\varphi}\cdot\frac{\partial}{\partial\varphi}\left[\sin\varphi\cdot\frac{\partial}{\partial\varphi}\left[\overbrace{R(r)}^{\text{const.}}\cdot Y(\theta,\varphi)\right]\right] + \frac{1}{\sin^2(\varphi)}\cdot\frac{\partial}{\partial\theta}\left[\frac{\partial}{\partial\theta}\left(\overbrace{R(r)}^{\text{const.}}\cdot Y(\theta,\varphi)\right)\right]$$

$$= R(r)\cdot\underbrace{\left\{\frac{1}{\sin\varphi}\cdot\frac{\partial}{\partial\varphi}\left[\sin\varphi\,\frac{\partial}{\partial\varphi}\left[Y(\theta,\varphi)\right]\right] + \frac{1}{\sin^2(\varphi)}\cdot\frac{\partial^2}{\partial\theta^2}\left[Y(\theta,\varphi)\right]\right\}}_{\text{Note } \Lambda_3 Y(\theta,\varphi)}$$

$$= R(r)\cdot\Lambda_3 Y(\theta,\varphi).$$

(5) $\frac{1}{r^2}\cdot(4) = \frac{1}{r^2}\cdot R(r)\cdot \Lambda_3 Y(\theta,\varphi)$

So now combining the results from (3) & (5) for Laplace eq-n in (3):

$$\frac{1}{r^2}\left[r^2\cdot R'(r)\cdot Y(\theta,\varphi)\right]'_r + \frac{1}{r^2}R(r)\cdot\Lambda_3 Y(\theta,\varphi) = 0$$

$$Y(\theta,\varphi)\cdot\frac{1}{r^2}\cdot\frac{\partial}{\partial r}\left[r^2\cdot R'(r)\right] + \frac{1}{r^2}\cdot R(r)\cdot\Lambda_3 Y(\theta,\varphi) = 0 \quad \Big\{ *r^2 \text{ of BS} \Rightarrow$$

$$Y(\theta,\varphi)\cdot\frac{\partial}{\partial r}\left[r^2\cdot R'(r)\right] + R(r)\cdot\Lambda_3 Y(\theta,\varphi) = 0 \Rightarrow$$

$$\frac{\frac{\partial}{\partial r}\left[r^2\cdot R'(r)\right]}{R(r)}\cdot\frac{Y(\theta,\varphi)}{Y(\theta,\varphi)} = -\frac{R(r)}{R(r)}\cdot\frac{\Lambda_3 Y(\theta,\varphi)}{Y(\theta,\varphi)} \Rightarrow$$

since we're considering $R(r)\neq 0$, $Y(\theta,\varphi)\neq 0$

$$\Rightarrow \underbrace{\frac{\left[r^2\cdot R'(r)\right]'}{R(r)}}_{(4)} = \underbrace{-\frac{\Lambda_3 Y}{Y}}_{(5)} = \mu = \text{const.} \in \mathbb{R}$$

[The only way (4) ≡ (5) is both ≡ $\mu \in \mathbb{R}$]

Hence by (4),

$$\frac{\left[r^2\cdot R'(r)\right]'}{R(r)} = \mu \Rightarrow \overbrace{\left[r^2\cdot R'(r)\right]' - \mu\cdot R(r) = 0}^{(6)}$$

And by (5)

$$-\frac{\Lambda_3 Y}{Y} = \mu \Rightarrow \overbrace{\Lambda_3 Y - \mu Y = 0}^{(7)}$$

(6) & (7) are the required eq-ns with $\mu\in\mathbb{R}$ & $\Lambda_3(\cdot)$ has been defined above.